Full Stack Developer Hiring Challenge

Advanced Expense Tracker with Categories (Next.js + FastAPI + SQL)

Problem Statement

Build a full-stack Expense Tracker with relational database design. Each expense must belong to a category. The system should support core CRUD operations and category-based grouping.

Time Limit

30 Minutes

Tech Stack Requirements

Frontend: Next.js

Backend: FastAPI

Database: SQL (SQLite preferred)

Backend Requirements (Detailed)

1. Category Management

- Create an API to add a new category

- Each category should have a unique id and name

- Duplicate category names should be avoided

```
POST /categories
Response:
{
  "id": 1,
  "name": "Food"
}
```

2. Expense Creation

- Create an API to add a new expense

- Each expense must be linked to a valid category

- Store title, amount, category_id, and timestamp

```
POST /expenses
Response:
```

```
{
  "id": 1,
  "title": "Lunch",
  "amount": 250.5,
  "category_id": 1,
  "created_at": "2026-04-14T12:00:00"
}
```

3. Fetch Expenses

- Create an API to fetch expenses list

- Return expenses with category name and created_at

```
GET /expenses
Response:
[
  {
    "id": 1,
    "title": "Lunch",
    "amount": 250.5,
    "category": "Food",
    "created_at": "2026-04-14T12:00:00"
  }
]
```

4. Category Summary

- Create an API to list categories

- Each category should include total number of expenses

```
GET /categories
Response:
[
  {
    "id": 1,
    "name": "Food",
    "total_expenses": 3
  }
]
```

5. Overall Summary

- Create an API to return overall statistics

- Include total amount, total count, and average

```
GET /summary
Response:
{
  "total": 430.5,
  "count": 2,
  "average": 215.25
}
```

Database Requirements (Detailed)

Table 1: categories

```
categories (
  id PRIMARY KEY,
  name TEXT
)
```

Table 2: expenses

```
expenses (
  id PRIMARY KEY,
  title TEXT,
  amount NUMERIC,
  category_id INTEGER,
  created_at DATETIME,
  FOREIGN KEY (category_id) REFERENCES categories(id)
)
```

Frontend Requirements (Detailed)

1. Category Section

- Input to create new category

- Display list of categories

2. Expense Form

- Input for title

- Input for amount

- Dropdown to select category

- Submit button to create expense

3. Expense List

- Display all expenses

- Show title, amount, category name, and date

4. Summary Section

- Display total amount

- Display total number of expenses

- Display average amount

5. Data Refresh

- After any action, update UI accordingly

Constraints

1. Category must exist before creating expense

2. Title cannot be empty

3. Amount must be positive

4. System should handle invalid inputs gracefully